

WEEK 1 (Day 4 & Day 5)

Patterns-

```
for i in range(1,6):  
    for j in range(1 , i+1):  
        print("*",end=" ")  
    print()
```

```
C:\Users\bhavn\OneDrive\Desktop\DataAn1\.venv\Scripts\python.exe  
*  
* *  
* * *  
* * * *  
* * * * *  
  
Process finished with exit code 0
```

```
for i in range(1,6):  
    for j in range(1 , i+1):  
        print(i,end=" ")  
    print()
```

```
C:\Users\bhavn\OneDrive\Desktop\DataAn1\.venv\Scripts\python.exe  
1  
2 2  
3 3 3  
4 4 4 4  
5 5 5 5 5  
  
Process finished with exit code 0
```

```
for i in range(1,8):
    for j in range(8,i,-1):
        print(i,end=" ")
    print()
```

C:\Users\bhavn\OneDrive\Desktop\DataAn1\.venv\Scripts\python.exe

```
1 1 1 1 1 1 1
2 2 2 2 2 2
3 3 3 3 3
4 4 4 4
5 5 5
6 6
7
```

Process finished with exit code 0

```
for i in range(1,11):
    for j in range(1,i+1):
        print(i*j,end=" ")
    print()
```

C:\Users\bhavn\OneDrive\Desktop\DataAn1\.venv\Scripts\python.exe

```
1
2 4
3 6 9
4 8 12 16
5 10 15 20 25
6 12 18 24 30 36
7 14 21 28 35 42 49
8 16 24 32 40 48 56 64
9 18 27 36 45 54 63 72 81
10 20 30 40 50 60 70 80 90 100
```

Process finished with exit code 0

String Manipulation in Python

String manipulation means performing operations on strings such as accessing, modifying, combining, searching, and formatting text.

A **string** is a sequence of characters enclosed in single (' ') or double (" ") quotes.

1. Accessing Characters (Indexing)

Characters can be accessed using index numbers.

```
text = "Python"
print(text[2])
print(text[5])
```

```
C:\Users\bhav\OneDrive\Desktop\DataAn1\
t
n

Process finished with exit code 0
```

2. String Slicing

Used to extract a part of a string.

```
text = "Data Analytics"
print(text[0:3])
print(text[7:14])
```

```
C:\Users\bhav\OneDrive\Desktop\DataAn1\.venv\Scripts\python.exe
Dat
alytics

Process finished with exit code 0
```

3. Concatenation (Joining Strings)

Combine two or more strings using +.

```
first = "Data"
second = "Analytics"
print(first + " " + second)
```

```
C:\Users\bhavn\OneDrive\Desktop\DataAn1\.venv\  
Data Analytics  
  
Process finished with exit code 0
```

4. String Length

```
a = "Python end to end project"  
print("The length of the given string is :",len(a))
```

```
C:\Users\bhavn\OneDrive\Desktop\DataAn1\.venv\Scripts\python.exe  
The length of the given string is : 25  
  
Process finished with exit code 0
```

5.Format

```
a = "My name is {}"  
print(a.format("Bhavneet"))
```

```
C:\Users\bhavn\OneDrive\Desktop\DataAn1\.venv\Scripts\python.exe  
My name is Bhavneet  
  
Process finished with exit code 0
```

6. Finding and Replacing

```
text = "Python Programming"  
  
print(text.find("Programming"))  
  
text = "Hello World"  
  
print(text.replace( old: "World", new: "Python"))
```

```
C:\Users\bhavn\OneDrive\Desktop\DataAn1\.venv\Scripts\python.exe  
7  
Hello Python  
  
Process finished with exit code 0
```

1. Write a program to get Fibonacci series up to 10 numbers.

```
prev = 0
next = 1
n = int(input("Enter a number :"))
print(prev,end=" ")
print(next,end=" ")
for i in range(1,n+1):
    sum = prev + next
    print(sum,end=" ")
    prev = next
    next = sum
print("\nAbove is the Fibonacci series...")
```

```
C:\Users\bhavnn\OneDrive\Desktop\DataAn1\.venv\Scripts\python.exe
Enter a number :10
0 1 1 2 3 5 8 13 21 34 55 89
Above is the Fibonacci series...

Process finished with exit code 0
```

2. Write a program to check if a number is prime or not.

```
n = int(input("Enter a number :"))
if n <= 1:
    print("Not a prime number")

for i in range(2,n):
    if n%i == 0:
        print("Not a prime number")
        break
    else:
        print("Prime number")
        break
```

```
Enter a number :4
Not a prime number

Process finished with exit code 0
```

```
Enter a number :37
Prime number

Process finished with exit code 0
```

3. Write a program to find a palindrome of integers.

```
str1 = input("Enter a string :")
s = 0
e = len(str1) - 1
while(s < e):
    if str1[s]== str1[e]:
        s += 1
        e -= 1
    else:
        print("Not a Palindrome")
        break
else:
    print("Palindrome")
```

```
C:\Users\bhav\OneDrive\Desktop\DataAn1\.venv\Scripts\python.exe
Enter a string :112211
Palindrome

Process finished with exit code 0
```

```
C:\Users\bhav\OneDrive\Desktop\DataAn1\.venv\Scripts\python.exe
Enter a string :1223678812
Not a Palindrome

Process finished with exit code 0
```

4. Write a program to create an area calculator.

```
print("*****AREA CALCULATOR*****")
print("Press 1 for Square")
print("Press 2 for Circle")
print("Press 3 for Triangle")
print("Press 4 for Rectangle")
choice = int(input("Enter your choice: "))
```

```
if choice == 1:
    a = int(input("Enter a number: "))
    print("Area of Square is : " , a**2)
if choice == 2:
    r = int(input("Enter radius :"))
    print("Area of Circle is : " , 3.14*r**2)
if choice == 3:
    b = int(input("Enter base :"))
    h = int(input("Enter height :"))
    print("Area of Triangle is : " , 0.5*b*h)
if choice == 4:
    l = int(input("Enter length :"))
    b = int(input("Enter breadth :"))
    print("Area of Rectangle is : " , l*b)
```

*****AREA CALCULATOR*****

Press 1 for Square

Press 2 for Circle

Press 3 for Triangle

Press 4 for Rectangle

Enter your choice: 2

Enter radius : 5

Area of Circle is : 78.5

Process finished with exit code 0